Expectation of $g(X)$

Motivation example: Suppose that discrete RV X has distribution:

x	$-\sqrt{2}$	-1	0	1	$\sqrt{2}$
$P_X(x)$	0.2	0.1	0.4	0.1	0.2

Let $Y = X^2$

y	0	1	2
$P_Y(y)$	0.4	$0.1 \times 2 = 0.2$	$0.2 \times 2 = 0.4$

$$E(Y) = \sum_{y=0}^{2} y \cdot P_Y(y) = 0 \times 0.4 + 1 \times 0.2 + 2 \times 0.4 = 1$$

$$= (-\sqrt{2})^2 \cdot 0.2 + (-1)^2 \cdot 0.1 + 0^2 \cdot 0.4 + 1^2 \cdot 0.1 + \sqrt{2}^2 \cdot 0.2$$

More generally,

$$E[g(X)] = \begin{cases} \sum_{\forall x} g(x)\, p_X(x) & \text{discrete} \\ \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx & \text{continuous} \end{cases}$$

Ex. Let

$$f_X(x) = \begin{cases} x(4-x)\cdot\frac{1}{9}, & x \in (0,3) \\ 0, & \text{o.w.} \end{cases}$$

Also, let $Y = \min(X, 1)$.

Find $E(Y)$.

Set $g(x) = \min(x, 1)$, thus $Y = g(X)$.

NOTE: g is NOT linear, thereby

$$E[g(X)] \neq g(E(X))$$

$$E(Y) = E(g(X))$$

$$= \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

$$= \int_0^3 \underbrace{\min(x, 1)}_{} \frac{x \cdot (4-x)}{9}\, dx$$

$$\hookrightarrow \begin{cases} x, & x \in (0,1) \\ 1, & x \in [1, 3) \end{cases}$$

$$= \int_0^1 x \cdot \frac{x(4-x)}{9}\, dx + \int_1^3 1 \cdot \frac{x(4-x)}{9}\, dx$$

$$= \frac{1}{9}\left[\int_0^1 4x^2 - x^3\, dx + \int_1^3 4x - x^2\, dx \right]$$

$$= 0.935185$$